

Practice Questions for Strong Duality

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1 Questions

1. According to strong duality, what is the relationship between the optimal objective function values of the primal and dual problems?
 - a. The optimal objective function value of the primal problem is always greater than or equal to the optimal objective function value of the dual problem.
 - b. The optimal objective function value of the primal problem is always less than or equal to the optimal objective function value of the dual problem.
 - c. The optimal objective function values of the primal and dual problems are always equal, provided both problems have optimal solutions.
 - d. There is no guaranteed relationship between the optimal objective function values of the primal and dual problems.
 - e. The optimal objective function value of the primal problem is always greater than the optimal objective function value of the dual problem.
2. The Klee-Minty problem highlights a worst-case scenario for the simplex method. What key characteristic of this problem leads to its exponential time complexity?
 - a. The presence of many constraints.
 - b. The large magnitude of the coefficients in the objective function.
 - c. The highly skewed and elongated geometry of its feasible region.
 - d. The use of a specific pivot rule.
 - e. The number of variables being very large.
3. In the context of strong duality, what can we conclude if the primal problem is unbounded?
 - a. The dual problem is also unbounded.
 - b. The dual problem is infeasible.
 - c. The dual problem has an optimal solution.
 - d. The dual problem is unbounded in the opposite direction.
 - e. There is no relationship between the primal and dual problems in this case.
4. Why is smoothed analysis considered a more realistic measure of the simplex method's performance than worst-case analysis?

- a. Smoothed analysis considers only average-case scenarios, ignoring worst-case instances.
 - b. Smoothed analysis accounts for the fact that real-world problems are rarely perfectly structured to trigger worst-case behavior.
 - c. Smoothed analysis focuses solely on the number of iterations, ignoring computational time per iteration.
 - d. Smoothed analysis is computationally simpler than worst-case analysis.
 - e. Smoothed analysis always provides polynomial bounds on the runtime of the simplex method.
5. In the context of strong duality, what is the relationship between the optimal objective function values of the primal and dual problems, assuming both problems have optimal solutions?
- a. The optimal objective function value of the primal problem is less than or equal to the optimal objective function value of the dual problem.
 - b. The optimal objective function value of the primal problem is greater than or equal to the optimal objective function value of the dual problem.
 - c. The optimal objective function values of the primal and dual problems are equal.
 - d. There is no relationship between the optimal objective function values of the primal and dual problems.
 - e. The optimal objective function value of the primal problem is the negative of the optimal objective function value of the dual problem.
6. The Klee-Minty problem is designed to demonstrate what characteristic of the simplex method?
- a. Its average-case polynomial time complexity.
 - b. Its ability to handle degenerate problems efficiently.
 - c. Its worst-case exponential time complexity.
 - d. Its effectiveness with large-scale problems.
 - e. Its insensitivity to the choice of pivot rule.
7. In the context of strong duality, what can we conclude if the primal problem is unbounded?
- a. The dual problem is also unbounded.
 - b. The dual problem is infeasible.
 - c. The dual problem has an optimal solution.
 - d. The dual problem is unbounded in the opposite direction.
 - e. There is no relationship between the primal and dual problem's solutions.
8. Why is smoothed analysis considered a more realistic measure of the simplex method's performance than worst-case analysis?
- a. Smoothed analysis considers only average-case scenarios, ignoring worst-case instances.
 - b. Smoothed analysis accounts for the fact that real-world problems are rarely perfectly structured to create worst-case scenarios.

- c. Smoothed analysis focuses solely on the number of iterations, ignoring computational time per iteration.
 - d. Smoothed analysis uses only the largest-coefficient rule, which is the most efficient pivot rule.
 - e. Smoothed analysis proves that the simplex method always runs in polynomial time.
9. According to strong duality, what is the relationship between the optimal objective function values of the primal and dual problems, assuming both problems have optimal solutions?
- a. The optimal objective function value of the primal problem is always greater than the optimal objective function value of the dual problem.
 - b. The optimal objective function value of the primal problem is always less than the optimal objective function value of the dual problem.
 - c. The optimal objective function values of the primal and dual problems are equal.
 - d. There is no relationship between the optimal objective function values of the primal and dual problems.
 - e. The relationship between the optimal objective function values depends on the specific problem instance.
10. According to strong duality, what is the relationship between the optimal objective function values of the primal and dual problems, assuming both problems have optimal solutions?
- a. The optimal objective function value of the primal problem is greater than or equal to the optimal objective function value of the dual problem.
 - b. The optimal objective function value of the primal problem is less than or equal to the optimal objective function value of the dual problem.
 - c. The optimal objective function values of the primal and dual problems are equal.
 - d. There is no relationship between the optimal objective function values of the primal and dual problems.
 - e. The optimal objective function value of the primal problem is the negative of the optimal objective function value of the dual problem.

2 Answers

1. According to strong duality, what is the relationship between the optimal objective function values of the primal and dual problems?
 - a. This describes weak duality, not strong duality. Weak duality states that the primal objective function value is always greater than or equal to the dual objective function value for any feasible solutions.
 - b. This describes weak duality, but reversed. Weak duality states that the primal objective function value is always greater than or equal to the dual objective function value for any feasible solutions.
 - c. Strong duality states that if both the primal and dual problems have feasible solutions, then their optimal objective function values are equal. This is a fundamental theorem in linear programming.
 - d. This is incorrect; strong duality establishes a precise relationship between the optimal objective function values under certain conditions.
 - e. This is incorrect; strong duality states that the optimal objective function values are equal, not that one is strictly greater than the other.
2. The Klee-Minty problem highlights a worst-case scenario for the simplex method. What key characteristic of this problem leads to its exponential time complexity?
 - a. While the number of constraints can influence runtime, it's the specific structure of the constraints and the resulting geometry that causes the exponential behavior in the Klee-Minty problem.
 - b. While large coefficients can affect the numerical stability of the simplex method, the exponential complexity of the Klee-Minty problem is primarily due to the geometry of the feasible region.
 - c. The Klee-Minty problem's feasible region is a highly skewed polytope. The simplex method, following certain pivot rules, can traverse a path along the boundary of this polytope that is exponentially long in the number of variables.
 - d. The choice of pivot rule is important, and different rules can lead to different numbers of iterations. However, the Klee-Minty problem demonstrates exponential complexity even with specific pivot rules, highlighting the inherent problem structure.
 - e. The number of variables is a factor, but the exponential complexity arises from the interaction between the number of variables and the specific geometry of the feasible region.
3. In the context of strong duality, what can we conclude if the primal problem is unbounded?
 - a. If the primal is unbounded, the dual cannot be unbounded. If both were unbounded, strong duality would be violated.
 - b. If the primal problem is unbounded (its objective function can be made arbitrarily large), then the dual problem must be infeasible. This is a direct consequence of strong duality.
 - c. If the primal is unbounded, the dual cannot have an optimal solution. Strong duality would be violated.
 - d. The dual problem's unboundedness would be in the same direction as the primal (minimizing instead of maximizing).
 - e. Strong duality provides a relationship between the primal and dual problems even in cases of unboundedness or infeasibility.

4. Why is smoothed analysis considered a more realistic measure of the simplex method's performance than worst-case analysis?
 - a. Smoothed analysis does not ignore worst-case scenarios; it incorporates them into a broader framework that considers perturbations of the input data.
 - b. Worst-case analysis focuses on the most unfavorable input, which is often artificially constructed and rarely encountered in practice. Smoothed analysis acknowledges that real-world problems have some randomness or imperfections, making the worst-case scenario less likely.
 - c. Smoothed analysis considers the overall runtime, which includes both the number of iterations and the computational cost per iteration.
 - d. Smoothed analysis is generally more complex than worst-case analysis because it involves probabilistic considerations.
 - e. Smoothed analysis provides probabilistic bounds, not guaranteed polynomial bounds, on the runtime of the simplex method. The bounds depend on the magnitude of the perturbations.
5. In the context of strong duality, what is the relationship between the optimal objective function values of the primal and dual problems, assuming both problems have optimal solutions?
 - a. This describes weak duality, which states that the objective function value of any feasible solution to the primal problem is less than or equal to the objective function value of any feasible solution to the dual problem. Strong duality is a stronger statement about the optimal solutions.
 - b. This is the reverse of weak duality and is incorrect in the context of strong duality.
 - c. Strong duality states that if both the primal and dual problems have optimal solutions, then their optimal objective function values are equal. This is a fundamental theorem in linear programming.
 - d. This is false; strong duality establishes a direct relationship between the optimal objective function values.
 - e. There's no reason to expect the optimal values to be negatives of each other. Strong duality states they are equal.
6. The Klee-Minty problem is designed to demonstrate what characteristic of the simplex method?
 - a. The Klee-Minty problem directly contradicts this; it shows the worst-case scenario, not the average case.
 - b. While the simplex method can struggle with degeneracy, the Klee-Minty problem's primary purpose is to illustrate exponential worst-case complexity, not specifically degeneracy issues.
 - c. The Klee-Minty problem is a specifically constructed linear program that forces the simplex method (with certain pivot rules) to take an exponential number of iterations to reach the optimal solution, thus demonstrating its worst-case exponential time complexity.
 - d. The Klee-Minty problem highlights the opposite; it shows a situation where the simplex method performs poorly.
 - e. The Klee-Minty problem's design is highly sensitive to the pivot rule used; different rules can lead to vastly different numbers of iterations.
7. In the context of strong duality, what can we conclude if the primal problem is unbounded?
 - a. If the primal is unbounded, the dual cannot be unbounded as well. They cannot both be unbounded simultaneously.

- b. If the primal problem is unbounded (its objective function can be made arbitrarily large), then the dual problem must be infeasible. This is a direct consequence of strong duality.
 - c. If the primal is unbounded, the dual cannot have an optimal solution. The dual must be infeasible.
 - d. The dual problem's unboundedness would be in the same direction (minimizing) if the primal is maximizing and unbounded.
 - e. Strong duality establishes a clear relationship between the solutions of the primal and dual problems.
8. Why is smoothed analysis considered a more realistic measure of the simplex method's performance than worst-case analysis?
- a. Smoothed analysis does not ignore worst-case scenarios; it aims to bridge the gap between worst-case and average-case analysis by considering slightly perturbed instances.
 - b. Smoothed analysis acknowledges that real-world linear programming problems are unlikely to be precisely structured to trigger the worst-case exponential behavior observed in problems like the Klee-Minty cube. It introduces small random perturbations to the input data, making the analysis more representative of real-world scenarios.
 - c. Smoothed analysis considers the overall computational complexity, not just the number of iterations.
 - d. Smoothed analysis does not restrict itself to a single pivot rule; it can be applied to various pivot rules.
 - e. Smoothed analysis does not prove polynomial time complexity in all cases; it shows that with small random perturbations, the simplex method exhibits polynomial behavior with high probability.
9. According to strong duality, what is the relationship between the optimal objective function values of the primal and dual problems, assuming both problems have optimal solutions?
- a. This describes weak duality for a maximization primal problem, not strong duality. Weak duality only provides an inequality, not equality.
 - b. This describes weak duality for a minimization primal problem, not strong duality. Weak duality only provides an inequality, not equality.
 - c. Strong duality states that if both the primal and dual problems have optimal solutions, then their optimal objective function values are equal. This is a fundamental theorem in linear programming.
 - d. This is incorrect; strong duality establishes a precise relationship between the optimal objective function values.
 - e. Strong duality provides a definitive relationship that holds regardless of the specific problem instance, provided both problems have optimal solutions.
10. According to strong duality, what is the relationship between the optimal objective function values of the primal and dual problems, assuming both problems have optimal solutions?
- a. This describes weak duality, not strong duality. Weak duality states that the objective function value of the primal problem is always greater than or equal to the objective function value of the dual problem for a maximization primal problem.
 - b. This describes weak duality, but reversed. Weak duality states that the objective function value of the primal problem is always greater than or equal to the objective function value of the dual problem for a maximization primal problem.

- c. Strong duality states that if both the primal and dual problems have optimal solutions, then their optimal objective function values are equal. This is a fundamental theorem in linear programming.
- d. This is incorrect. Strong duality establishes a precise relationship between the optimal objective function values of the primal and dual problems.
- e. This is incorrect. There is no such relationship between the optimal objective function values of the primal and dual problems.